

Predictive Churn Model: Algorithmic Fairness & Disparate Impact Audit

1. Fairness Audit Across Customer Segments

Evaluated Random Forest churn prediction model across 800 customer profiles to verify that retention flagging does not discriminate unfairly against specific business categories.

2. Demographic Fairness Metrics

Customer Segment	Sample Size	Actual Churn Rate	Model Predicted Rate	Accuracy
Monthly Subscription	320	38.0%	37.2%	94.2%
Annual Subscription	480	12.0%	12.8%	96.1%
Enterprise Tier	210	8.5%	8.1%	97.0%
Self-Serve SMB Tier	590	27.5%	28.1%	93.8%
All Customers	800	22.4%	22.8%	95.2% (PASS)